Multiscale Context-Aware Ensemble Deep KELM for Efficient Hyperspectral Image Classification

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***Abstract*— Recently, multiscale spatial features have been widely utilized to improve the hyperspectral image (HSI) classification performance. However, fixed-size neighborhood involving the contextual information probably leads to misclassifications, especially for the boundary pixels. Additionally, it has been demonstrated that deep neural network (DNN) is practical to extract representative features for the classification tasks. Nevertheless, under the condition of high dimensionality versus small sample sizes, DNN tends to be over-fitting and it is generally time-consuming due to the deep-level feature learning process. To alleviate the aforementioned issues, we propose a multiscale context-aware ensemble deep kernel extreme learning machine (MSC-EDKELM) for efficient HSI classification. First, the scene of the HSI data set is over-segmented in multiscale via using the adaptive superpixel segmentation technique. Second, superpixel pattern (SP) and attentional neighboring superpixel pattern (ANSP) are generated by leveraging the superpixel maps, which can automatically comprise local and global contextual information, respectively. Afterward, an ensemble deep kernel extreme learning machine (EDKELM) is presented to investigate the deep-level characteristics in the SP and ANSP. Finally, the category of each pixel is accurately determined by the decision fusion and weighted output layer fusion strategy. Experimental results on four real-world HSI data sets demonstrate that the proposed frameworks outperform some classic and state-of-the-art methods with high computational efficiency, which can be employed to serve real-time applications.**

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***Index Terms*— Deep kernel extreme learning machine (DKELM), HSI classification, multiscale spatial feature, superpixel segmentation.**

I. Introduction

BENEFITING from wealthy spatial and spectral information, hyperspectral imagery (HSI) has been employed in various applications increasingly, such as mineral resource exploitation, precision agriculture, and environmental earth sciences [1]–[4]. In practice, HSI classification is critical in these applications, which aims to accurately assign predefined labels to the individual pixels of the scene. However, due to the intrinsic characteristics of HSI, such as high dimensionality against limited labeled training samples and spatial discrepancy of spectral signatures, HSI classification with high accuracy is still a challenging task [5]–[7].

Over the last few decades, many researchers devoted their efforts to addressing these problems and plenty of related approaches have been investigated, such as random forest [8], Bayesian models [9], and kernel methods [10]. In particular, support vector machine (SVM) as a milestone has shown excellent classification performance even with limited labeled samples [11]. Whereas, these traditional methods can only exploit shallow spectral–spatial features, which is generally underrepresented. To cope with this defect, the deep neural networks (DNNs) are utilized and have become the mainstream in recent years [12], [13]. Compared with the aforementioned classification models, the most distinct character of DNNs is to learn high-level features in an end-to-end hierarchical framework, which is more likely to the pattern recognition procedure of humans. In this manner, the extracted deep features are more refined and representative, which are proved to be beneficial for classification problems.

Furthermore, due to its local receptive fields and shared weights, convolutional neural networks (CNNs) establish a dominant position for dealing with the visual-based tasks [14], [15]. In this context, Hu *et al.* [16] directly employed the spectral curves to develop a 1-D-CNN to categorize varying materials in the HSI and yield impressive results. However, 1-D-CNN handles each pixel (i.e., sample) independently and fails to take advantage of the correlations among different image pixels. To release this limitation, spatial contextual information was naturally introduced, which is actually of great significance in HSI classification [17]. For example, Li *et al.* [18] presented a 3-D-CNN model to simultaneously explore spectral–spatial features contained in spatially

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neighboring pixels, which obtained encouraging performance. In spite of being successful, 1) without sufficient training samples, the traditional DNNs face an over-fitting problem in HSI classification and 2) it generally exhausts much time to well-train such hierarchical deep architectures to learn the effective high-level features.

To address the first shortcoming, Li *et al.* [19] proposed a CNN-based on pixel-pair features (CNN-PPF) through modifying and relabeling the available labeled samples. As a result, the training sample size was largely augmented so as to acquire promising classification results. Specifically, extracting more representative and discriminative spectral–spatial features can also alleviate this small sample size (SSS) problem. Hence, similar to 3-D-CNN, under the assumption that neighboring pixels have high probability to be the same object, most models adopt the 3-D neighborhood with the identical label to the central pixel as input to learn representative spectral–spatial features in an end-to-end manner [20]–[22]. However, the pixels contained in one fixed-size patch may not belong to the same category, especially for the boundary areas across categories. Consequently, it brings bias to the classifier and lead to undesirable misclassifications. To counteract this problem, some works exploited multiscale convolutional filter bank to capture more refined spatial features at different levels [23], [24], other researches utilized multiscale patches to leverage the compensate spatial information, such as multiscale CNN (MSCNN) [25] and diverse-region CNN (DR-CNN) [26]. Nonetheless, the predefined-shape neighbors still cannot perfectly resolve this issue.

Fortunately, superpixel segmentation as an adaptive neighborhood system can be introduced. Generally, similar properties of pixels in local area can be reflected in one superpixel, such as color, intensity, and texture [27]. Incorporating HSI classification, Fang *et al.* [28] utilized multiple kernels to exploit spectral–spatial information contained in the superpixels. Moreover, from under-segmentation to over-segmentation, different number of superpixel regions can generate multiscale features of one pixel. For instance, a multiscale superpixel-based approach was proposed to integrate the classification results acquired from each scale [29]. In [30], a multiscale superpixel segmentation was applied to model the classes distribution relying on spatial information. However, they only exploit the feature extracted from each local superpixel region, which is under-segmentation or over-segmentation. It obviously cannot capture the contextual information of the pixels precisely. To this end, we propose a context-aware multiscale perception module to automatically characterize the local–global contextual information at different scales.

To overcome the second problem, the computationally efficient extreme learning machine (ELM) [31] is taken into consideration. Specifically, the only parameters of ELM to be trained are output weights, which can be easily calculated by a smallest norm least-squares solution but not the commonly used complicated gradient-based back-propagation (BP). Thus, it is much more competent in efficiency compared with the deep CNNs. In fact, plentiful works about ELM have already been investigated concerning hyperspectral classification and achieved remarkable contributions [32]–[35]. However, the number of hidden nodes as hyperparameter is pretty hard to determine, and different results appear even with the same nodes because of the randomly generated weights and bias. Besides, ELM is easy to be overfitting. Subsequently, the kernel version ELM (KELM) [36] was proposed to circumvent these deficiencies, in which the hidden layer of the ELM is substituted by a kernel function. Although KELM has been applied to HSI classification and obtained satisfactory results [37], [38], as mentioned earlier, the feature representation capacity is relatively limited of such shallow networks. Technically, deriving inspiration from the multilayer perception (MLP) theory, Huang extended ELM to multilayer ELM (ML-ELM) through stacking ELM-based autoencoder (ELM-AE) for deep feature representation [39], [40]. From the view of deep learning, ML-ELM with deep layers can obviously further extract deep robust and abstract features. Nevertheless, manually adjusting the number of hidden nodes in each layer is an uncertainty for training efficiency and generalization [41]. To get rid of this disadvantage, sparse representation is adopted in [42], which introduced an excellent multilayer sparse ELM (MSELM) model for landcover identification in HSIs. Moreover, deep kernel ELM (DKELM) is developed for HSI classification through KELM-based autoencoder (KELM-AE), which achieves optimal generalization and obtains encouraging classification results [43].

Inspired by the aforementioned approaches, we propose a multiscale context-aware ensemble deep kernel extreme learning machine (MSC-EDKELM) for efficient and accurate HSI classification. First, the sequential numbers of superpixels are adaptively determined assisted by the inherent spatial size and spatial resolution of the data set. Then, multiscale superpixel segmentation maps are generated through using Entropy Rate Segmentation (ERS) [44] method performed on the first principal component (PC) [45] of the data set. Second, the original HSI is integrated with segmentation maps via context-aware multiscale perception module such that the pixel-level data can be represented by superpixel pattern (SP) and attentional neighboring superpixel pattern (ANSP). Finally, these flexibly designed patterns along with the original spectral features are fed into the ensemble DKELM, which is composed of several independent DKELM to hierarchically capture the deep and representative spectral–spatial features. Eventually, the accurate classification maps are obtained through decision fusion strategy (DFS) and weighted output layer fusion strategy (WOLFS).

In summary, the main contributions of the proposed MSC-EDKELM are as follows:

1) Based on the adaptive multiscale superpixel segmentation, we propose a context-aware multiscale perception module to flexibly exploit the local–global contextual information and acquire diverse feature representation.
2) The automatically designed SP and ANSP along with the original spectral features are deeply investigated by the proposed EDKELM. Moreover, both DFS and WOLFS are employed to produce the eventual accurate results.
3) Compared with the competitive approaches in performance, the proposed method is relatively efficient and

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exhausts less time to obtain satisfactory classification results, which can be used for real-time applications.

4) The proposed algorithm is evaluated on four benchmarks, particularly, a new hyperspectral data set is utilized, which is recently collected by advanced hyperspectral imager (AHSI) mounted on China's Gaofen5 satellite platform.

The remains of this are organized as follows. Section II briefly introduces the related works. Section III describes our proposed method in detail. Abundant experiments are performed and the classification results are analyzed in Section IV. The conclusions are drawn in Section V.

II. Related Works

A. Superpixel Segmentation

In recent years, plentiful superpixel segmentation algorithms have been widely investigated in computer vision filed. In particular, ERS is one of the most promising methods, which is inspired by the graph theory to achieve the homogeneous clustering [44]. To be specific, it can be summarized in the following two steps.

First, specify the desired number of superpixels, S, over the whole image. And then construct the graph $G = (V, E)$, where V is the collection of vertices in accord with pixels, and E is the edge set computed by the Gaussian decreasing function through using pixel distance. Second, the main part of ERS is to seek an optimal subset $Q \subseteq E$, which is able to divide the graph G into S connected subgraphs. Formally, the segmentation task can be transferred to solve the following optimization:

$$\arg\max_{Q} H(Q) + \gamma B(Q) \tag{1}$$

where $H(Q)$ refers to the constraint of entropy rate to make homogeneous clusters. $B(Q)$ represents the balance term to facilitate the clusters to have a similar spatial size. $\gamma \geq 0$ is a coefficient to adjust the influence of the balance term.

B. Kernel Extreme Learning Machine

Assume that we have N training samples denoted as $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^{N} \in R^{N\times L}$, and the corresponding labels are $\mathbf{Y} = \{\mathbf{y}_i\}_{i=1}^{N} \in R^{N\times C}$, where C is the number of classes. For each $\mathbf{y}_i = [y_1, \ldots, y_c, \ldots, y_C]$, if $y_c = 1$ and other elements equal to -1, then $\mathbf{x}_i$ belongs to the cth class. Specifically, the illustration of ELM is depicted in Fig. 1.

Then, for an arbitrary example $\mathbf{x}_i$, the output function of ELM with D hidden nodes can be represented as

$$f_{\mathrm{ELM}}(\mathbf{x}_i) = \sum_{j=1}^{D} \boldsymbol{\varphi}_j h\left(< \mathbf{w}_j, \mathbf{x}_i > + b_j\right) = \mathbf{y}_i \tag{2}$$

where $\mathbf{w}_j$ and b_j is the input weight and bias of the jth hidden node connected to the input features, respectively, and they are randomly assigned. $\boldsymbol{\varphi}_j$ is the output weight of the jth hidden node connected to the output nodes. $< \mathbf{w}_j, \mathbf{x}_i >$ denotes the dot product of $\mathbf{w}_j$ and $\mathbf{x}_i$. Here, h is the sigmoid activation function. We can find that the hidden layer is nonlinear while the output layer is linear.

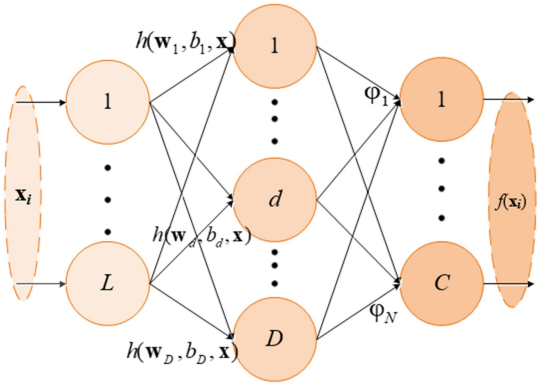

Fig. 1. Illustration of ELM.

Alternatively, 2 can be expressed in matrix form

$$\mathrm{H}\boldsymbol{\psi} = \mathbf{Y} \tag{3}$$

where $\boldsymbol{\psi} = [\varphi_1, \ldots, \varphi_D] \in R^{D\times C}$, and $\mathbf{H} = \begin{bmatrix} \mathbf{h}(\mathbf{x}_1) \\ \vdots \\ \mathbf{h}(\mathbf{x}_N) \end{bmatrix}$ refers to the output matrix of the hidden layer which can be obtained through

$$\mathbf{H} = \begin{bmatrix} h(\mathbf{w}_1 \cdot \mathbf{x}_1 + b_1) & \cdots & h(\mathbf{w}_D \cdot \mathbf{x}_1 + b_D) \\ \vdots & \ddots & \vdots \\ h(\mathbf{w}_1 \cdot \mathbf{x}_N + b_1) & \cdots & h(\mathbf{w}_D \cdot \mathbf{x}_N + b_D) \end{bmatrix}_{N\times D}. \tag{4}$$

Then, $\boldsymbol{\psi}$ can be calculated by using a smallest norm least-squares solution

$$\boldsymbol{\psi} = \mathbf{H}^{\dagger}\mathbf{Y} = \mathbf{H}^{\mathrm{T}}\left(\frac{I}{\rho} + \mathbf{H}\mathbf{H}^{\mathrm{T}}\right)^{-1}\mathbf{Y} \tag{5}$$

where ρ is a regularization parameter to enhance the stability and generalization ability. Finally, the output function of ELM model can be formulated as

$$f_{\mathrm{ELM}}(\mathbf{x}_i) = h(\mathbf{x}_i)\mathbf{H}^{\mathrm{T}}\left(\frac{\mathbf{I}}{\rho} + \mathbf{H}\mathbf{H}^{\mathrm{T}}\right)^{-1}\mathbf{Y}. \tag{6}$$

Using kernel tricks to substitute the explicit activation function $h(\mathbf{x}_i)$, the ELM can be extended to KELM. Specifically, let $\boldsymbol{\Omega} = \mathbf{H}\mathbf{H}^{\mathrm{T}}$, where $\boldsymbol{\Omega}_{i,j} = K(\mathbf{x}_i, \mathbf{x}_j)$ and K is a kernel function satisfying the Mercer's conditions. Then the output of KELM can be expressed as

$$f_{\mathrm{KELM}}(\mathbf{x}_i) = \begin{bmatrix} \boldsymbol{\Omega}_{i,1} \\ \vdots \\ \boldsymbol{\Omega}_{i,N} \end{bmatrix}\left(\frac{\mathbf{I}}{\rho} + \boldsymbol{\Omega}\right)^{-1}. \tag{7}$$

Benefited from the implicit kernel mapping, there is no need to predefine the number of hidden nodes, which equals to the number of training samples in KELM. Besides, the generalization ability of the model is further improved.

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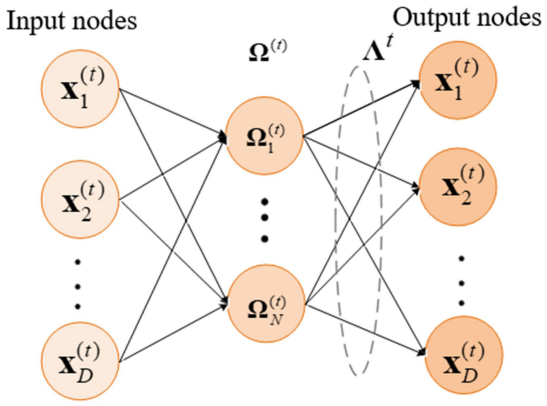

Fig. 2. Illustration of KELM-AE.

C. KELM Autoencoder

Majority of research studies have demonstrated that the deep abstract and refined features are more significant for the classification task. The representing unsupervised deep networks include stacked autoencoders (SAE) and deep belief networks (DBNs). Inspired by these instructive works, KELM with a single hidden layer can be developed to be an effective deep network through autoencoder.

To be more specific, the structure of KELM-AE is depicted in Fig. 2. As shown in Fig. 2, the input matrix $\mathbf{X}^{(t)}$ is transformed into a hidden kernel matrix $\mathbf{\Omega}^{(t)}$ by a kernel function $K^{(t)}(\mathbf{x}_i, \mathbf{x}_j)$. Denote the tth transformation matrix as $\mathbf{\Lambda}^{(t)}$, then we have

$$\mathbf{\Omega}^{(t)}\mathbf{\Lambda}^{(t)} = \mathbf{X}^{(t)}. \tag{8}$$

Next, $\mathbf{\Lambda}^{(t)}$ can be solved via

$$\mathbf{\Lambda}^{(t)} = \left(\frac{\mathbf{I}}{\rho} + \mathbf{\Omega}^{(t)}\right)^{-1}\mathbf{X}^{(t)}. \tag{9}$$

Finally, the input data of next AE can be computed by

$$\mathbf{X}^{(t+1)} = h(\mathbf{X}^{(t)}(\mathbf{\Lambda}^{(t)})^{\mathrm{T}}) \tag{10}$$

where h is a linear or nonlinear activation function.

III. Proposed Method

The main architecture of our proposed MSC-EDKELM for HSI classification is exhibited in Fig. 3. We can see that it is basically composed of three parts: multiscale superpixel segmentation, context-aware multiscale perception and ensemble DKELM. Concretely, Fig. 3(a) displays the DFS while Fig. 3(b) depicts the WOLFS. In the following, these three portions of the framework are elaborated in detail.

A. Multiscale Superpixel Segmentation for HSI

In complex land covers, since the objects are usually in different geometric appearances, single scale segmentation is incapable of properly characterizing the abundant spatial attributes. Accordingly, we involve multiscale superpixel segmentation into a unified framework to exhaustively exploit the rich contextual information. Specifically, we choose ERS method for over-segmentation since it performs well to obtain more compact segmentation maps with an efficient computation. In particular, we adopt the first PC of the HSI as the segmentation target, which contains the most spectral–spatial information. Compared with operating on original HSI, it can not only enhance the separability of pixels, but also boost the segmentation process with less data volume [46].

The numbers of superpixels in multiscale are the vital parameters in the proposed method which can be determined by various strategies [47]–[49]. Here, we employ a heuristic method to adaptively determine the superpixel numbers considering the inherent properties of the HSIs, i.e., spatial size and spatial resolution. Concretely, the number of superpixels at scale m is calculated by

$$S_m = \mathrm{floor}\left(\frac{H \times W}{\mathrm{floor}(100 \times 0.7^{\sqrt{\mathrm{res}}}) \times m}\right), \quad m = \{1, 2, \ldots, M\} \tag{11}$$

where floor means the value down to the nearest integer. $H \times W$ equals to the spatial size and res refers to the spatial resolution. Meanwhile, M represents the number of considered scales. Please note that, in the subsequent context-aware multiscale perception module, both superpixels and adjacent superpixels are simultaneously exploited at each scale, thus the spectral–spatial information can be fully investigated with less scales. Hence, we adjust 11 to enlarge the interval between two adjacent scales by replacing m to 2^{m-1}. For instance, the illustration of multiscale superpixel maps on Indian Pines data set are depicted in Fig. 4. Consequently, 3-D multiscale superpixels are obtained through masking these 2-D maps onto the aligned HSI data set.

B. Context-Aware Multiscale Perception Module

In order to fully explore the spectral–spatial features, we present a context-aware multiscale perception module, which is composed of the SP and ANSP.

Given a HSI data set referred to $\mathbf{D} = (\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_{H\times W}) \in R^{H\times W\times L}$, where H and W are spatial dimensions and L is the length of the spectral dimension. In the process of multiscale superpixel over-segmentation, each superpixel is supposed to contain landcover only belonging to one category since it is homogeneous. Under this assumption, the SP can be obtained by the average of pixels involved in one superpixel. If $\mathbf{x}_i$ is a training sample, then the SP of $\mathbf{x}_i$ at scale m can be represented as

$$\mathbf{x}_{\mathrm{im}}^{\mathrm{SP}} = \frac{1}{|\mathbf{\Psi}_{\mathrm{im}}|}\sum_{j=1}^{|\mathbf{\Psi}_{\mathrm{im}}|}\mathbf{p}_j \tag{12}$$

in which $\mathbf{\Psi}_{\mathrm{im}}$ denotes the superpixel where $\mathbf{x}_i$ locates at scale m and $|\cdot|$ is used to count the number of pixels in $\mathbf{\Psi}_{\mathrm{im}}$.

We can find that SP is capable of gathering adaptive local spatial information of adjacent pixels. Then, to further precisely capture the global contextual information, we introduce the ANSP, which is formulated as

$$\mathbf{x}_{\mathrm{im}}^{\mathrm{ANSP}} = \frac{\sum_j \mathbf{W}_{i,j}\mathbf{z}_j}{\sum_j \mathbf{W}_{i,j}}, \quad \mathbf{W}_{i,j} = e^{-\alpha d(\mathbf{x}_i, \mathbf{z}_j)} \tag{13}$$

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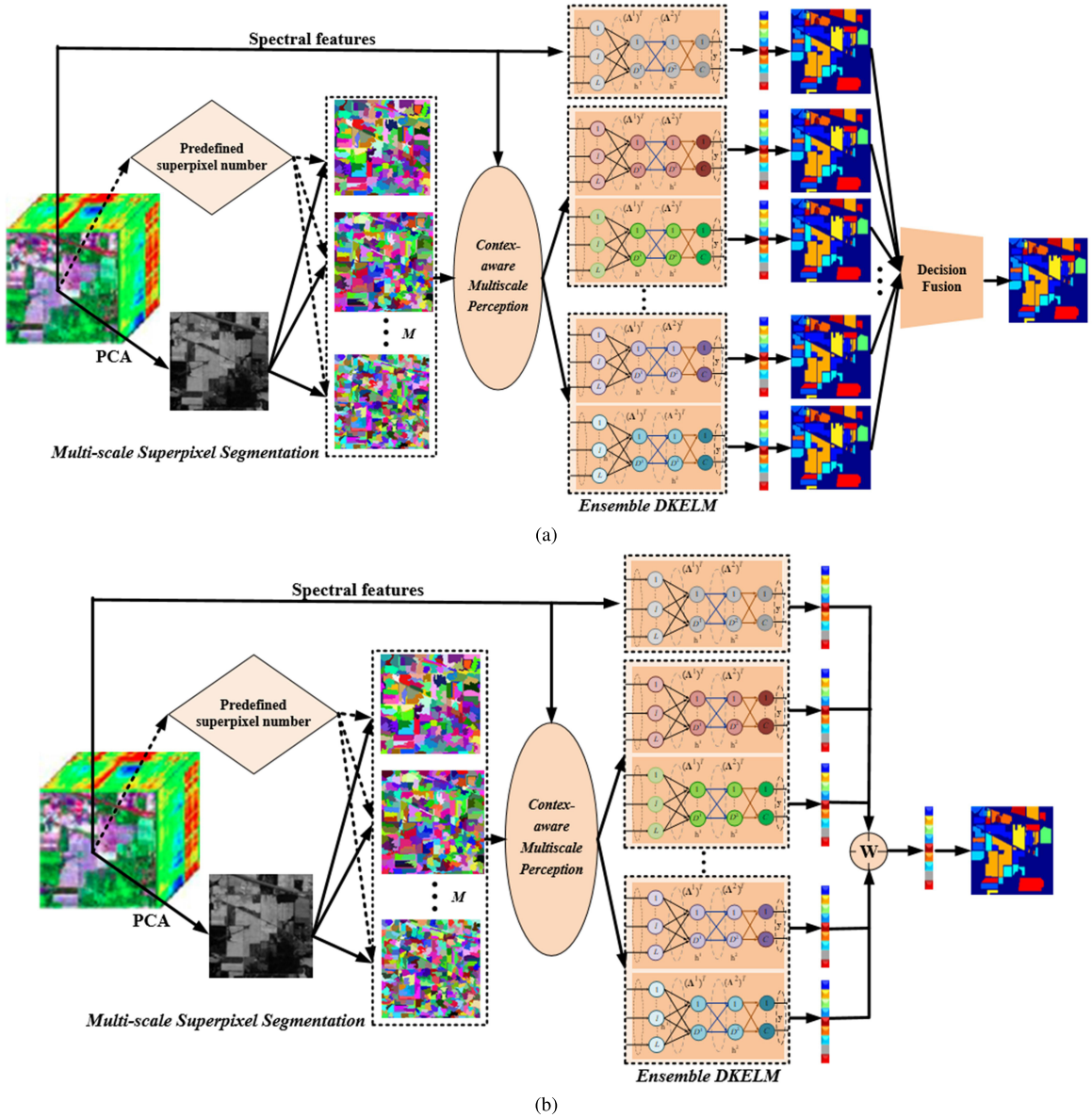
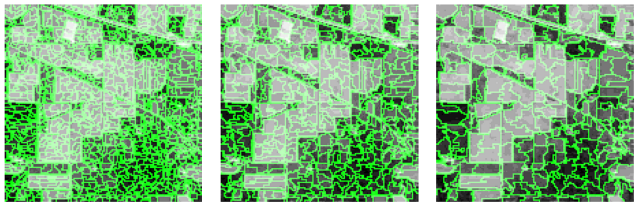


Fig. 3. Illustration of the proposed multiscale context-aware ensemble deep KELM frameworks. (a) DFS. (b) WOLFS.



(a) (b) (c)

Fig. 4. Illustration of multiscale 2-D superpixel maps on Indian Pines image scene. (a) $S_1 = 1051$. (b) $S_2 = 525$. (c) $S_3 = 262$.

where $\mathbf{z}_j$ is the anchor point of the jth neighboring supepixel of $\mathbf{\Psi}_{\text{im}}$, and it is obtained via averaging the spectral vectors in the corresponding superpixel. $\mathbf{W}_{i,j}$ refers to the attention score of $\mathbf{z}_j$ to $\mathbf{x}_i$, which is a decreasing function of the spectral angle distance between $\mathbf{x}_i$ and $\mathbf{z}_j$ denoted as $d(\mathbf{x}_i, \mathbf{z}_j)$. α is a steepness parameter and $d(\mathbf{x}_i, \mathbf{z}_j)$ is calculated by

$$d(\mathbf{x}_i, \mathbf{z}_j) = \arccos \frac{\mathbf{x}_i^{\mathrm{T}} \mathbf{z}_j}{\sqrt{\mathbf{x}_i^{\mathrm{T}} \mathbf{x}_i} \sqrt{\mathbf{z}_j^{\mathrm{T}} \mathbf{z}_j}}. \tag{14}$$

Fig. 5 shows the illustration of ANSP, where $j \in [1, 2, \ldots, 5]$ in $\mathbf{W}_{i,j}$ since there are five adjacent superpixels of $\mathbf{\Psi}_{\text{im}}$. It can be observed that ANSP can adaptively capture long-range global contextual information based on the prior spectral correlations.

C. Ensemble DKELM

Based on the abovementioned analysis, we can find that multiscale SP and ANSP can adequately integrate the spectral–spatial information contained in the HSI. From other view, that procedure can be regarded as a data augmentation strategy to generate diverse spectral–spatial features. For further

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Fig. 5. Illustration of ANSP of the training sample $\mathbf{x}_i$. Various colors represent different superpixels.

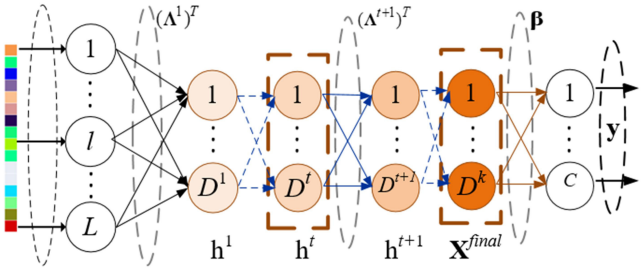

Fig. 6. Main structure of DKELM.

exploiting these features efficiently, we propose an ensemble DKELM framework. Specifically, the SP and ANSP in M scales along with the original spectral features are fed into $2M+1$ independent DKELM classifiers, and then the $2M+1$ output are comprehensively processed by DFS or WOLFS, and finally we can obtain the predicted label of each pixel. In the following, the DKELM algorithm, DFS and WOLFS are described in detail, respectively.

1) DKELM: In order to extract deep and effective features, deep kernel extreme learning machine (DKELM) can be developed in a self-taught manner through stacking the KELM-AEs. Fig. 6 shows the main structure of DKELM. In Fig. 6, $\mathbf{\Lambda}^{(1)}$ refers to the transformation matrix of the first layer in DKELM, which is obtained by the KELM-AE with the input data. $\mathbf{\Lambda}^{(t+1)}$ represents the transformation matrix of the $(t+1)$th hidden layer h^{t+1}, which is acquired by KELM-AE with the tth hidden layer h^t as input nodes. Actually, each couple of $\mathbf{\Lambda}^{(t)}$ and $h^{(t)}$ can be calculated by 9 and 10, respectively. In the end, the final data representation $\mathbf{X}^{\text{final}}$ is obtained and then it is used to train a KELM classifier as

$$\mathbf{\Omega}_{\text{final}}\boldsymbol{\beta} = \mathbf{Y} \tag{15}$$

where $\mathbf{\Omega}_{\text{final}}$ is mapped from $\mathbf{X}_{\text{final}}$ via using the kernel function. Here, we use the Radial Basis Function (RBF), which is referred to $K(\mathbf{x}_i,\mathbf{x}_j) = \exp(-\left\|\mathbf{x}_i - \mathbf{x}_j\right\|/2\gamma^2)$. Then, the final output weight $\boldsymbol{\beta}$ is solved by

$$\boldsymbol{\beta} = \left(\frac{\mathbf{I}}{\rho} + \mathbf{\Omega}_{\text{final}}\right)^{-1}\mathbf{Y}. \tag{16}$$

Different from the greedy layer-wise training of deep learning based on gradient descent, the hidden layers of DKELM are trained in a forward approach. The transformation matrices of DKELM are all obtained through the smallest least-square solution without any fine-tuning. Therefore, DKELM can capture the deep effective features more efficiently than the normal DNN frameworks.

2) DFS: As observed in Fig. 3, we can get $2M+1$ prediction maps from $2M+1$ DKELM classifiers. Then, DFS applies majority voting to fuse the output maps to obtain the final classification result. That is, for a sample $\mathbf{x}$, the estimated label is determined by

$$C(\mathbf{x}) = \underset{c=\{1,2,\ldots,C\}}{\arg\max}\ M_{\mathbf{x}}(c), \quad M_{\mathbf{x}}(c) = \sum_{k=1}^{2M+1}\mathbf{I}(C(\mathbf{x}^k) = c) \tag{17}$$

where C($\mathbf{x}$) refers to the final predicted label of $\mathbf{x}$. $M_{\mathbf{x}}(c)$ counts the votes that pixel $\mathbf{x}$ is categorized to class c. $\mathbf{I}$ denotes the indicator function and $C(\mathbf{x}^k)$ represents the judged label of $\mathbf{x}$ from the kth DKELM.

3) WOLFS: The DFS simply treats each scale of SP and ANSP in each DKELM to be equal regardless of their different importance. To conquer this defect, we develop a WOLFS, which can be formulated as

$$C(\mathbf{x}) = \underset{c=\{1,2,\ldots,C\}}{\arg\max}\ \hat{\mathbf{y}}, \quad \hat{\mathbf{y}} = (\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_C) = \sum_{k=1}^{2M+1}\mathbf{f}^k \odot \hat{\mathbf{y}}^k \tag{18}$$

where $\hat{\mathbf{y}}^k$ represents the output layer of the kth DKELM and $\odot$ denotes the element-wise product operator. The cth element of $\mathbf{f}^k \in R^C$ depends on the cth classification accuracy obtained by the kth DKELM on the training data. To be more specific, the f_c^k can be derived as

$$\begin{aligned} u_c^k &= \frac{\sum_{n=1}^{N}\sum_{\arg\max \mathbf{y}_n^k = c}\mathbf{I}(\arg\max \hat{\mathbf{y}}_n^k = \arg\max \mathbf{y}_n^k)}{\sum_{n=1}^{N}\mathbf{I}(\arg\max \mathbf{y}_n^k = c)} \\ f_c^k &= \frac{u_c^k + 10^{-5}}{\sum_{i=1}^{2M+1}u_c^i + 10^{-5}} \end{aligned} \tag{19}$$

where u_c^k is the cth class accuracy of the kth DKELM classifier.

IV. Experimental Results and Analysis

In this section, the performance of the proposed MSC-EDKELM is evaluated on four real-world data sets, including the Indian Pines, the University of Pavia, the Kennedy Space Center (KSC), and the Yancheng data set. Some classic and state-of-the-art classification models are employed as compared algorithms, such as SVM-RBF [11], KELM [37], DKELM [43], CNN-PPF [19], MSCNN [25], DR-CNN [26]. Furthermore, we adopt class-specified accuracy (CA), average accuracy (AA), overall accuracy (OA), kappa coefficient (k) as criteria metrics.

A. Experiment Data Sets

1) Indian Pines: The first Indian Pines data set was captured by the Airborne/Infrared Imaging Spectrometer (AVIRIS) over Northwestern Indiana. This image is composed of 145 × 145 pixels with a relatively low spatial resolution as 20 m. The AVIRIS collects 220 bands in the wavelength range

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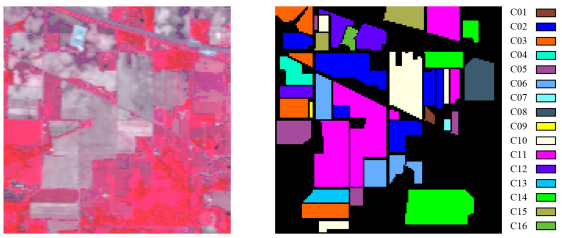


(a) (b)

Fig. 7. Composite false-color image and groundtruth for Indian Pines data set.

TABLE I

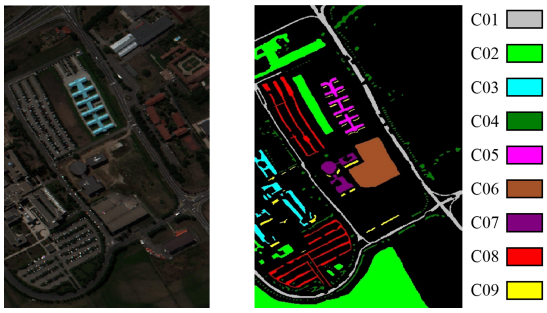
LAND COVER TYPES WITH THE NUMBER OF TRAINING AND TEST SAMPLES FOR INDIAN PINES DATA SET

ID	Land Cover Class	Training	Test
C01	Alfalfa	15	31
C02	Corn-notill	15	1413
C03	Corn-mintill	15	815
C04	Corn	15	222
C05	Grass-pasture	15	468
C06	Grass-trees	15	715
C07	Grass-pasture-mowed	15	13
C08	Hay-windrowed	15	463
C09	Oats	15	5
C10	Soybean-notill	15	957
C11	Soybean-mintill	15	2440
C12	Soybean-clean	15	578
C13	Wheat	15	190
C14	Woods	15	1250
C15	Buildings-Grass-Trees-Drives	15	371
C16	Stone-Steel-Towers	15	78
	Total	240	10009

of 0.4–2.5 μm. In our experiments, there are 200 bands left after removing the noisy and water absorption bands. The composite false color image and the groundtruth are shown in Fig. 7(a) and (b), respectively. Table I gives the detailed number of training and test samples for 16 land cover classes. We can find that the number of annotated samples are imbalance and in small size for some classes, e.g., class C07 and C09. Therefore, Indian Pines is a challenging HSI data set which is widely used to develop new classification algorithms.

2) University of Pavia: The second benchmark data set, University of Pavia, was captured by the Reflective Optics Spectrometer (ROSIS) sensor over Pavia, northern Italy. This image consists of 610 $\times$ 340 pixels and the geometric resolution is 1.3 m. After discarding the noisy bands, 103 channels are remained covering the range from 0.43 to 0.86 μm. Fig. 8 shows the composite false color image and its groundtruth. A total number of 42776 samples from nine landcovers are labeled, of which the training and test portions are clearly listed in Table II.

3) Kennedy Space Center: The third real-world HSI data set is KSC, which was acquired by the AVIRIS sensor over the KSC, FL, USA. The spatial size of this image is 512 $\times$ 614, and the spatial resolution is 18 m. Only 176 bands ranging from 0.4 to 2.5 μm are used in our experiments



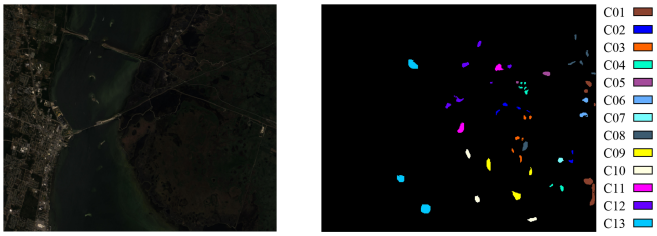
(a) (b)

Fig. 8. Composite false-color image and groundtruth for University of Pavia data set.

TABLE II

LAND COVER TYPES WITH THE NUMBER OF TRAINING AND TEST SAMPLES FOR UNIVERSITY OF PAVIA DATA SET

ID	Land Cover Class	Training	Test
C01	Asphalt	30	6601
C02	Meadows	30	18619
C03	Gravel	30	2069
C04	Trees	30	3034
C05	Painted metal sheets	30	1315
C06	Bare Soil	30	4999
C07	Bitumen	30	1300
C08	Self-Blocking Bricks	30	3652
C09	Shadows	30	917
	Total	270	42506



(a) (b)

Fig. 9. Composite false-color image and groundtruth for KSC data set.

after the removal of noisy and water absorption bands. There are 5211 labeled samples from 13 landcovers. Fig. 9(a) and (b) depicts its false-color composite image and groundtruth, respectively, while Table III concretely describes the training and test samples of each class.

4) Yancheng: The last experiment data set named Yancheng was generated by the AHSI, which is mounted on the platform of China's Gaofen5 satellite [50]. This image scene was collected on April 4, 2019 over the Yancheng, Jiangsu province, China. The AHSI sensor collects as much as 330 spectral channels ranging from 0.4 to 2.5 μm by the visible and near-infrared (VNIR) and short-wave infrared (SWIR) detectors. However, a total of 64 bands are removed, i.e., 1–14, 88, 129–133 for VNIR and 42–51, 96–115, 119–123, 171–173, 175–180 for SWIR. Thus, the preserved 266 bands are utilized in the experiments. Particularly, this data set covers a large scene of 1175 $\times$ 585 pixels and each of them has a spatial resolution of 30 m. The composite false color image and

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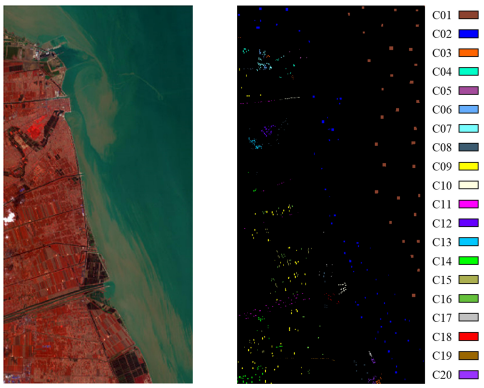
TABLE III

LAND COVER TYPES WITH THE NUMBER OF TRAINING AND TEST SAMPLES FOR KSC DATA SET

ID	Land Cover Class	Training	Test
C01	Srub	30	731
C02	Willow swamp	30	213
C03	CP hammock	30	226
C04	Slash pine	30	222
C05	Oak/Broadleaf	30	131
C06	Hardwood	30	199
C07	Swamp	30	75
C08	Graminoid	30	401
C09	Spartina marsh	30	490
C10	Cattail marsh	30	374
C11	Salt marsh	30	389
C12	Mud flats	30	473
C13	Water	30	897
	Total	390	4821



(a) (b)

Fig. 10. Composite false-color image and groundtruth for Yancheng data set.

the groundtruth are displayed in Fig. 10. Explicitly, there are 20 types of labeled materials in the image, of which the training and test sample numbers are provided in Table IV.

B. Parameter Tuning

From Tables I to IV, we can observe that 15 labeled samples from each class are randomly selected as the training set for Indian Pines and Yancheng data sets, and 30 labeled examples are randomly chose in the training process for University of Pavia and KSC data sets. Correspondingly, the remained labeled samples are taken to test the proposed network. It is worth to note that 10% samples of the training data are randomly selected as validation set to determine the hyperparameters of the proposed MSC-EDKELM.

Specifically, there are several user-specified parameters need to be tuned, including the number of scales denoted as M for SP and ANSP, number of kernel layers D, regularization parameters ρ, and kernel function parameters γ for each individual DKELM. To simplify the process of parameter tuning, values of D, ρ, and $\boldsymbol{\gamma}$ for each independent DKELM in our proposed MSC-EDKELM keep the same. According to the validation performance, the scale number M and the

TABLE IV

LAND COVER TYPES WITH THE NUMBER OF TRAINING AND TEST SAMPLES FOR YANCHENG DATA SET

ID	Land Cover Class	Training	Test
C01	Sea	15	2380
C02	Off shore water	15	1290
C03	Salina	15	95
C04	Pond	15	118
C05	Marsh	15	6
C06	Spartina	15	68
C07	Mudflate	15	203
C08	Aquculture	15	213
C09	Paddy	15	817
C10	Eatuarine water	15	157
C11	River	15	202
C12	Woodland	15	141
C13	Bare land	15	139
C14	Architecture	15	345
C15	Fallow land	15	219
C16	Dry land	15	189
C17	Suaeda	15	70
C18	Mariculyure	15	20
C19	Ditch	15	23
C20	Reed	15	117
	Total	300	6812

kernel layer number D are both empirically set to 3 for all experimental data sets. The influence of these two parameters will be further analyzed in Section IV-F. Besides, the superpixel numbers in three scales are calculated by 11, which are, in turn, {1051, 525, 262}, {3142, 1571, 785}, {14289, 7144, 3572}, and {49098, 24549, 12274} for the four experiment data set. In addition, since the proposed network includes three kernel layers, the regularization parameters can be denoted as $\rho = \{\rho_1, \rho_2, \rho_3\}$, and the parameters used in the RBF kernel function are $\gamma = \{\gamma_1, \gamma_2, \gamma_3\}$.

In the simulations, the value of $\{\rho_1, \rho_2, \rho_3\}$ are chosen from the candidate set {1e-4, 1e-3, 1e-2, 1e-1, 1, 1e1, 1e2, 1e3, 1e4}. To make full use of our proposed framework, we utilize the grid search algorithm to find the optimal parameters. Fig. 11 depicts the classification performance with various of combination of ρ_1, ρ_2, and ρ_3 for all experiment data sets. In Fig. 11, the two horizontal coordinate axes express ρ_1 and ρ_2, and the vertical coordinate axis represents ρ_3. The color bars mean the classification accuracy obtained via the different sets of values. From Fig. 11, we can observe that, better classification performance is achieved if ρ_1 is small while ρ_2 and ρ_3 are in large values. The black circles pointed by the red arrow with best classification accuracy indicate the values we finally chosen. Here, we list the selected values employed in the four data sets: {1e-2, 1e4, 1e2}, {1e-1, 1e3, 1e4}, {1, 1e4, 1e3} and {1e-1, 1e4, 1e2}. In a similar way, the kernel parameters are sought from {1e-1, 1, 1e1, 1, 1e1, 1e2, 1e3, 1e4, 1e5, 1e6, 1e7} and finally assigned to {1e2, 1e5, 1e7} for all of the four data sets.

C. Classification Performance

To demonstrate the superior performance of the proposed MSC-EDKELM, some other well-known classifiers are employed as comparisons. Normally, the classic SVM

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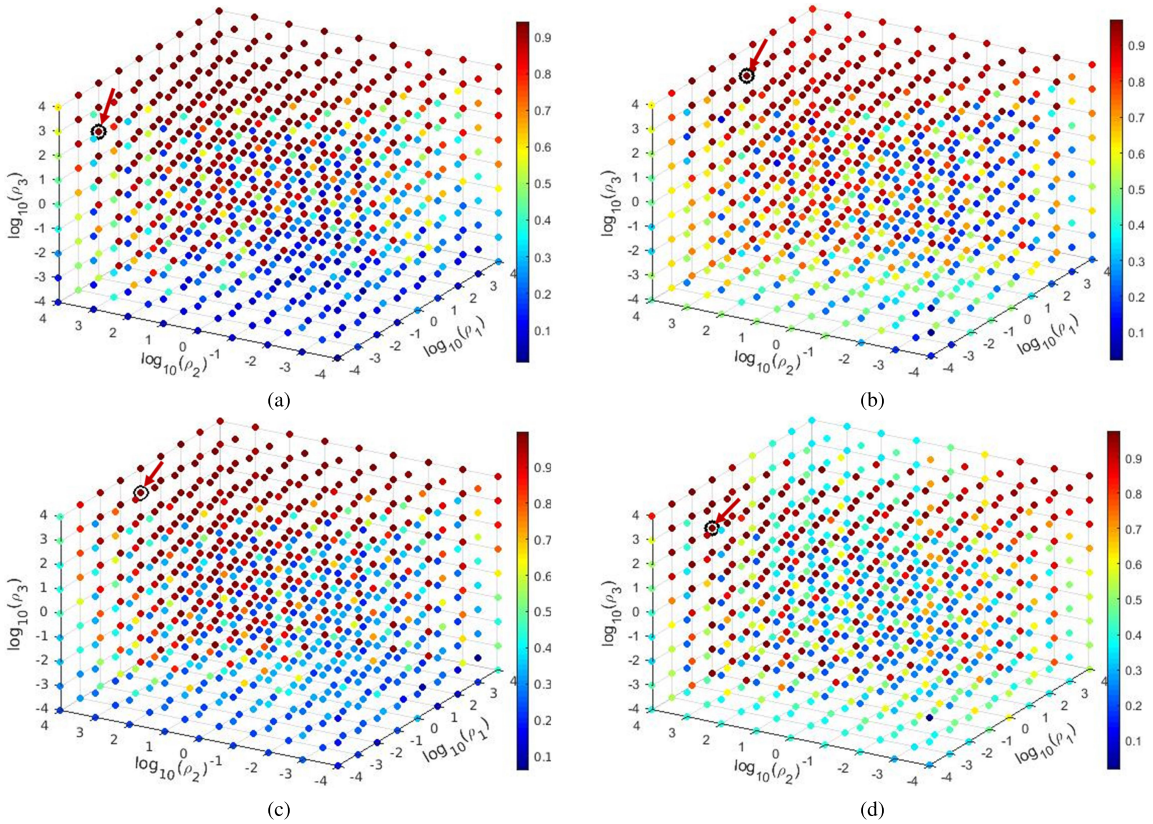

Fig. 11. Regularization parameters tuning for the proposed MSC-EDKELM. (a) Indian Pines. (b) University of Pavia. (c) KSC. (d) Yancheng. Best zoomed-in view.

with RBF kernel (SVM-RBF) is employed as the baseline method. Meanwhile, we compare the proposed framework with the highly related algorithms, i.e., KELM and DKELM. Besides, we adopt three state-of-the-art deep CNN-based spectral–spatial approaches, i.e., CNN-PPF, MSCNN, DR-CNN, of which the first method realizes data augmentation through using pixel-pair features and the last two methods also utilize multiscale spatial features. Considering the implements of the compared methods, the optimal hyperparameters of SVM-RBF and KELM are tuned via five-fold cross validation while the remained methods are configured as the original papers. In order to ensure the reliability of the experimental results, all of these methods are repeated ten times on each data set, and the mean accuracy along with standard deviations are reported from Tables V to VIII. For convenience of comparison, the highest accuracy of each row is highlighted in bold.

Table V depicts the quantitative experimental results for various methods on Indian Pines data set. It is observed that the proposed MSC-EDKELM outperforms the compared algorithms by a substantial margin in terms of AA, OA, kappa coefficient. In addition, the result obtained by DKELM is better than that of KELM due to the extraction of depth refined features. Besides, the spectral–spatial CNN-based methods outperform the SVM-RBF, KELM, and DKELM algorithms, which only leverage the spectral features. Furthermore, it is worth to notice that the MSC-EDKELM with WOLFS is superior to DFS, and the standard deviations are also very small. It implies that the WOLFS can make better use of the complementary local-global spatial context features and it is more stable and suitable than DFS on Indian Pines.

The classification accuracies for University of Pavia data set are summarized in Table VI. Our proposed frameworks still have optimal performance, although CNN-PPF, MSCNN and DR-CNN are satisfactory HSI classification algorithms. Specifically, the OA of MSC-EDKELM-WOLFS is 13.19%, 4.61%, and 2.20% higher than CNN-PPF, MSCNN, and DR-CNN, respectively. The reason that MSC-EDKELM achieves prominent results is mainly attributed to the context-aware perception mechanism, which can adaptively capture the variations of contextual distribution around objects. Namely, MSC-EDKELM can effectively perceive and handle the image scene with many irregular class boundaries, such as the University of Pavia data set. In addition, taking different importance of the ensemble classifiers into account, the well-designed WOLFS once again works better than the DFS.

Table VII provides the classification accuracies acquired by different methods on KSC data set. It is found that the entire performance is better than that of Indian Pines and University of Pavia data sets. The main reason lies in the aspect that KSC has higher spatial resolution than Indian Pines and less data noise than University of Pavia data set, which is more preferable for accurate classification. Meanwhile, our proposed MSC-EDKELM methods achieve the top-level performance compared with other algorithms. Particularly, there is no misclassification appears for ten of the 13 categories. Moreover,

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TABLE V
CLASSIFICATION ACCURACY OF DIFFERENT ALGORITHMS FOR INDIAN PINES DATA SET WITH FIFTEEN TRAINING SAMPLES PER CLASS

ID	SVM-RBF	KELM	DKELM	CNN-PPF	MSCNN	DR-CNN	MSC-EDKELM-DFS	MSC-EDKELM-WOFLS
C01	87.10±3.29	90.32±1.15	93.55±1.09	90.32±1.37	93.88±0.88	**97.23±0.55**	96.77±0.15	96.77±0.15
C02	48.12±2.76	45.15±2.35	47.49±5.79	59.52±2.75	77.63±1.41	80.97±1.67	**94.90±0.13**	93.49±0.27
C03	43.31±2.85	37.79±5.61	52.39±1.66	76.20±1.69	86.75±1.79	92.02±0.16	93.13±1.39	**95.21±0.54**
C04	79.73±1.98	81.08±0.65	81.53±1.39	92.34±0.87	99.55±0.23	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
C05	82.69±1.01	88.46±1.97	85.90±2.96	91.88±0.59	93.59±0.47	94.02±0.55	**94.23±0.46**	93.38±0.59
C06	79.72±2.06	90.35±0.35	88.53±2.33	93.57±1.84	98.11±0.41	**99.86±0.08**	99.16±0.19	99.72±0.04
C07	84.62±0.78	84.62±2.64	84.62±0.54	76.92±3.76	94.12±0.34	**95.37±0.77**	92.31±0.33	92.31±0.33
C08	81.21±1.43	80.78±1.86	78.19±1.35	96.33±0.81	97.34±0.85	**100.00±0.00**	99.78±0.37	99.78±0.26
C09	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
C10	61.34±2.59	61.96±3.79	62.17±2.46	68.13±2.58	81.25±2.33	83.31±1.62	91.64±0.69	**91.95±0.51**
C11	48.16±3.94	54.34±3.51	59.59±1.23	57.50±3.24	75.69±1.87	85.00±0.78	84.39±0.67	**87.30±1.03**
C12	45.67±5.63	48.96±2.30	48.62±1.92	78.20±1.95	86.33±1.29	97.23±0.16	95.16±1.19	**97.92±0.07**
C13	92.11±0.68	96.84±0.23	95.79±0.55	97.89±0.27	99.89±0.06	**100.00±0.00**	99.47±0.12	99.47±0.12
C14	60.16±2.51	66.08±1.98	67.68±1.32	85.12±1.36	96.08±0.11	**99.44±0.12**	99.04±0.36	99.12±0.21
C15	58.49±3.66	59.03±2.55	53.10±3.43	77.63±2.57	96.50±0.33	99.46±0.27	**100.00±0.00**	99.73±0.11
C16	89.74±1.41	87.18±1.04	89.74±0.67	91.03±0.23	**100.00±0.00**	92.31±1.07	**100.00±0.00**	**100.00±0.00**
AA (%)	71.39±1.86	73.31±1.19	74.31±1.26	83.29±0.69	92.29±0.12	94.76±0.36	96.25±0.31	**96.64±0.25**
OA (%)	58.21±2.39	60.98±2.21	63.40±1.07	73.64±1.51	84.16±1.31	89.98±0.86	93.35±0.65	**94.22±0.39**
$k \times 100$	53.18±2.88	56.24±1.85	58.89±0.89	70.37±1.66	82.10±1.45	88.63±0.77	92.43±0.71	**93.41±0.21**

TABLE VI
CLASSIFICATION ACCURACY OF DIFFERENT ALGORITHMS FOR UNIVERSITY OF PAVIA DATA SET WITH THIRTY TRAINING SAMPLES PER CLASS

ID	SVM-RBF	KELM	DKELM	CNN-PPF	MSCNN	DR-CNN	MSC-EDKELM-DFS	MSC-EDKELM-WOFLS
C01	73.31±2.55	42.40±5.31	73.93±2.16	91.56±1.32	94.06±1.43	**98.44±0.66**	87.45±1.62	88.74±1.59
C02	75.38±3.41	83.97±0.35	80.46±1.31	78.64±1.76	92.71±1.98	94.46±0.75	98.98±0.13	**99.65±0.06**
C03	75.35±1.44	76.66±1.22	78.40±0.98	87.48±1.15	96.18±1.22	95.41±1.19	99.80±0.03	**99.86±0.12**
C04	87.77±1.01	95.57±0.78	95.25±1.17	94.20±1.04	82.96±2.53	97.07±0.11	95.65±0.13	**97.69±0.35**
C05	99.92±0.02	99.62±0.17	99.62±0.13	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.82±0.13	99.77±0.10
C06	61.61±3.29	62.19±1.37	51.11±3.59	74.13±2.59	**100.00±0.00**	96.70±0.43	94.66±1.24	99.48±0.29
C07	85.69±1.89	79.92±2.31	76.85±1.88	94.85±1.32	99.46±0.25	97.15±0.27	99.67±0.04	**99.69±0.13**
C08	73.69±1.35	68.54±1.79	69.11±1.04	88.34±1.99	89.10±0.79	84.61±2.91	**97.41±1.34**	96.66±0.87
C09	98.91±0.22	97.60±0.55	96.18±1.39	**100.00±0.00**	61.18±2.49	**100.00±0.00**	95.76±0.46	88.00±2.63
AA (%)	81.29±1.36	78.62±1.27	80.10±1.01	89.91±1.36	90.63±1.44	95.98±0.26	96.58±0.39	**96.62±0.37**
OA (%)	75.76±1.23	74.88±1.44	76.80±1.76	84.11±1.18	92.69±0.92	95.10±0.57	96.43±0.22	**97.30±0.12**
$k \times 100$	68.93±2.34	67.44±1.67	69.81±0.65	79.55±0.86	90.42±0.39	93.55±0.49	95.26±0.43	**96.42±0.31**

the accuracy of MSC-EDKELM with DFS is still slightly lower than that of the MSC-EDKELM-WOLFS.

Table VIII presents the classification performance achieved by all of the considered methods. First, due to higher spectral resolution, the narrow bands of the Yancheng data set contain more clear divergences of different objects. Therefore, the overall classification results of this image are quite good among the four experimental data sets. Second, it is shown that MSC-EDKELM can obtain the best accuracies compared with other methods, which again demonstrates the strength of our proposed frameworks. Similarly, the WOLFS performs better than DFS in MSC-EDKELM on Yancheng data set.

The visual performance comparison of different methods on the experimental data sets are depicted from Figs. 12–15. The classification maps of each data set is in accordance with Tables V–VIII, respectively. For convenience to compare, the corresponding composite false-color image and groundtruth are also shown together. It is observed that the maps acquired by the proposed MSC-EDKELM-WOLFS are closest to the groundtruth. This indicates that there are the least misclassifications than other compared methods. Due to the lack use of spatial contextual information, the classification maps of SVM-RBF, KELM and DKELM suffer from salt-and-pepper noises within certain regions. In contrast, maps of MSC-EDKELM-WOLFS are the clearest and smoothest profited by the precise utilization of the contextual information, which is achieved through using the proposed context-aware multiscale perception module. Furthermore, to illustrate the power of the proposed method to handle the boundary pixels, the border regions of interest are highlighted in red circles for Indian Pines and University of Pavia data sets. We can easily find that the boundary regions of the proposed methods are much clearer and more compact than those of other methods.

D. Influence of Different Size of Training Samples

The experimental results from Tables V to VIII have fully demonstrated the excellence of our proposed method. In order to further evaluate the generalization ability of our proposed frameworks, we investigate the classification performance of

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TABLE VII
CLASSIFICATION ACCURACY OF DIFFERENT ALGORITHMS FOR KSC DATA SET WITH THIRTY TRAINING SAMPLES PER CLASS

ID	SVM-RBF	KELM	DKELM	CNN-PPF	MSCNN	DR-CNN	MSC-EDKELM-DFS	MSC-EDKELM-WOFLS
C01	85.77±1.23	92.34±0.68	91.52±1.19	99.59±0.13	85.64±1.35	**100.00±0.00**	99.45±0.13	89.06±1.13
C02	81.22±1.45	89.67±1.17	92.49±0.78	76.53±1.20	99.53±0.35	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
C03	80.09±1.78	84.51±0.88	86.28±0.55	87.17±0.79	95.58±0.78	98.37±0.31	99.56±0.10	**100.00±0.00**
C04	68.92±1.97	54.50±2.65	67.12±2.76	93.24±0.31	78.83±2.22	67.36±3.35	97.75±0.64	**98.20±0.53**
C05	74.81±1.03	82.44±0.53	80.15±1.43	95.42±0.58	**100.00±0.00**	83.44±1.28	97.71±0.77	**100.00±0.00**
C06	80.90±1.37	82.41±0.86	76.88±1.29	76.88±1.74	**100.00±0.00**	96.80±1.74	**100.00±0.00**	**100.00±0.00**
C07	94.67±0.35	85.33±1.24	94.67±0.27	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
C08	90.27±0.12	82.54±0.66	89.53±0.34	83.04±1.06	85.29±1.34	97.39±0.55	98.75±0.55	**100.00±0.00**
C09	91.22±0.26	98.57±0.21	95.10±1.22	92.04±0.79	82.65±2.12	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
C10	87.7±1.34	89.3±0.76	93.85±0.32	**100.00±0.00**	**100.00±0.00**	92.64±2.12	95.19±0.68	**100.00±0.00**
C11	96.92±0.13	94.34±0.45	95.63±0.89	**100.00±0.00**	**100.00±0.00**	99.51±0.32	**100.00±0.00**	**100.00±0.00**
C12	95.98±0.15	85.20±1.54	90.91±1.77	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	90.06±1.37	99.58±0.13
C13	98.22±0.24	98.55±0.33	98.55±0.46	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	96.54±1.56	**100.00±0.00**
AA (%)	86.67±0.37	86.13±0.56	88.67±0.89	91.80±0.59	94.42±0.18	95.04±0.11	98.08±0.34	**99.80±0.05**
OA (%)	89.46±0.54	89.52±0.63	91.29±0.43	93.09±0.40	93.63±0.62	96.91±0.19	97.64±0.56	**99.81±0.09**
$k \times 100$	88.24±0.27	88.29±0.49	90.26±0.38	92.34±0.52	92.90±0.44	96.55±0.13	97.35±0.11	**99.79±0.12**

TABLE VIII
CLASSIFICATION ACCURACY OF DIFFERENT ALGORITHMS FOR YANCHENG DATA SET WITH FIFTEEN TRAINING SAMPLES PER CLASS

ID	SVM-RBF	KELM	DKELM	CNN-PPF	MSCNN	DR-CNN	MSC-EDKELM-DFS	MSC-EDKELM-WOFLS
C01	85.80±1.87	93.45±1.01	96.39±1.36	96.01±1.77	95.76±1.34	96.52±1.05	**98.45±1.14**	98.28±1.21
C02	99.46±0.21	96.74±1.37	99.15±0.14	98.84±0.72	99.33±0.34	**100.00±0.00**	99.46±0.35	99.46±0.35
C03	90.53±0.55	75.79±2.33	85.26±0.68	**98.95±0.59**	97.89±0.65	93.68±1.22	97.89±0.68	97.89±0.68
C04	58.47±3.47	70.34±1.25	78.81±2.34	97.46±0.33	98.66±0.82	98.98±0.35	97.46±1.21	**100.00±0.00**
C05	33.33±5.82	83.33±2.36	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
C06	85.29±3.48	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
C07	71.92±0.57	81.28±0.98	90.64±1.46	**91.13±0.35**	74.53±3.49	76.58±2.25	90.64±1.38	89.66±1.69
C08	75.59±1.38	83.10±1.35	89.20±0.78	93.43±0.46	95.68±0.79	**99.53±0.07**	90.14±1.29	98.59±0.54
C09	80.54±1.26	96.33±0.66	98.16±0.69	97.67±0.71	93.27±1.25	90.70±2.43	**98.29±0.78**	96.70±1.51
C10	83.44±0.64	77.07±1.58	85.99±1.33	91.08±1.22	**95.18±1.23**	94.73±0.55	85.99±2.12	93.63±1.01
C11	64.85±1.78	58.42±3.64	65.84±2.51	**98.02±0.31**	88.61±1.20	78.71±1.39	80.20±1.50	84.65±1.45
C12	94.33±1.37	98.58±0.99	98.58±0.78	94.33±1.88	98.33±0.42	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
C13	83.45±1.21	89.93±1.46	99.28±0.15	**100.00±0.00**	91.39±0.42	96.49±1.55	**100.00±0.00**	97.84±1.20
C14	36.23±4.39	73.04±2.59	82.90±1.31	57.97±4.35	93.04±1.31	**98.84±0.40**	73.33±3.01	84.35±2.55
C15	80.82±1.66	65.75±4.95	76.26±3.49	93.15±0.46	94.98±1.08	92.24±1.33	**95.43±1.51**	93.61±1.49
C16	53.97±1.80	63.49±2.91	79.89±2.84	**99.47±0.08**	69.48±4.28	88.84±1.39	93.65±0.91	84.13±1.33
C17	70.00±1.98	78.57±2.30	94.29±2.55	95.71±0.29	95.71±1.39	95.71±1.37	88.57±2.68	**100.00±0.00**
C18	70.00±0.77	55.00±4.79	65.00±3.40	90.00±0.24	**97.33±0.69**	90.61±1.51	85.00±3.57	90.00±2.28
C19	69.57±3.49	91.30±2.56	**100.00±0.00**	86.96±1.71	88.61±1.56	90.57±1.76	**100.00±0.00**	91.30±1.57
C20	79.49±2.96	95.73±1.37	96.58±0.79	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
AA (%)	73.35±2.65	81.36±2.98	89.11±2.01	94.01±0.37	93.38±0.65	94.16±0.86	93.72±0.77	**95.00±0.61**
OA (%)	82.09±2.14	88.77±1.15	93.38±1.01	94.72±0.29	94.10±0.78	95.39±0.63	95.74±0.54	**96.37±0.38**
$k \times 100$	78.62±1.29	86.43±1.39	91.95±0.55	93.59±0.88	92.84±0.40	94.37±0.31	94.81±0.19	**95.58±0.11**

MSC-EDKELM and the other competitors under different numbers of labeled examples. Taking SSS into consideration, the training numbers are assigned to be relatively small. Specifically, the labeled samples of Indian Pines and Yancheng data sets are set from 3 to 15 with an interval of 3, and set from 10 to 30 with an interval of 5 for University of Pavia and KSC data sets. Meanwhile, the other experimental settings are kept same to Tables V–VIII. Here, each experiment is repeated ten times, and the mean OA are reported in Fig. 16. It is evident that the classification accuracy grows with the increase of labeled data for all methods, and our proposed MSC-EDKELM-WOLFS is always in the first place. Particularly, not like most of the compared methods, the OA of the MSC-EDKELM-WOLFS varies within a small range. This is attributed to the fact that context-aware multiscale perception module flexibly captures local and global spatial information, which is fully exploited by the followed DKELM. Moreover, the reasonable WOLFS is able to take the advantage of each individual DKELM classifier. Consequently, deep representative features are extracted and aggregated, which guarantee the proposed frameworks to be more stable to limited training samples.

E. Performance Under Controlled Random Sampling Scheme

In addition, we adopt the controlled random sampling scheme (CRSS) [51] to confirm the superiority the proposed frameworks. Specifically, 10% labeled samples are selected

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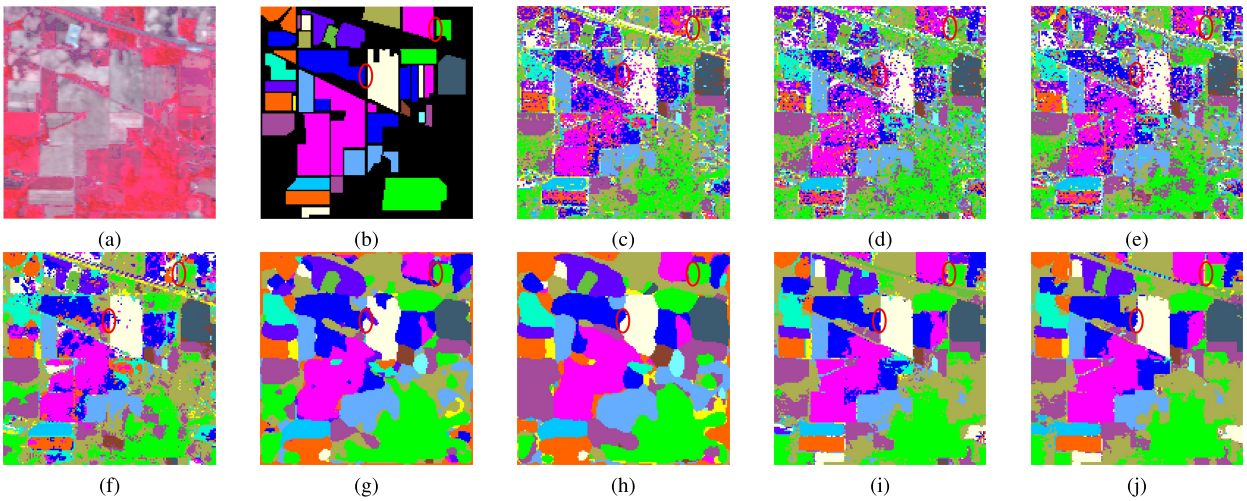

Fig. 12. Classification maps obtained via various algorithms for Indian Pines data set. (a) False-color image. (b) Groundtruth. (c) SVM-RBF. (d) KELM. (e) DKELM. (f) CNN-PPF. (g) MSCNN. (h) DR-CNN. (i) MSC-EDKELM-DFS. (j) MSC-EDKELM-WOLFS. Best zoomed-in view.

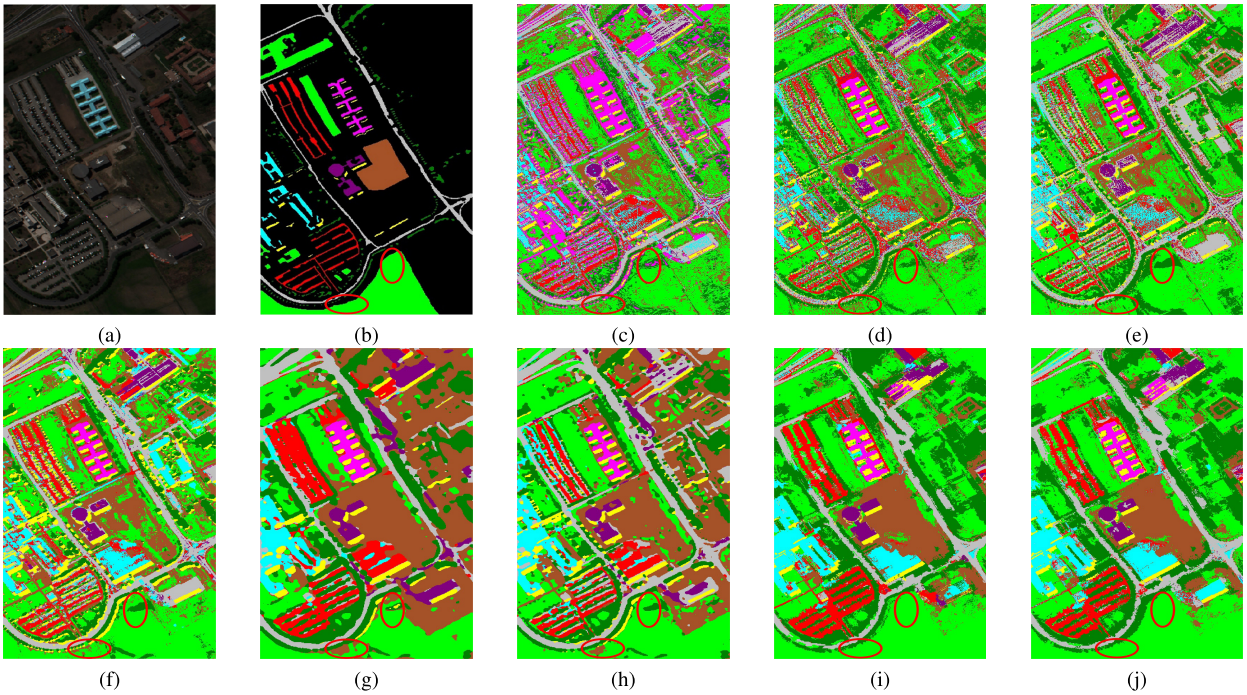

Fig. 13. Classification maps obtained via various algorithms for University of Pavia data set. (a) False-color image. (b) Groundtruth. (c) SVM-RBF. (d) KELM. (e) DKELM. (f) CNN-PPF. (g) MSCNN. (h) DR-CNN. (i) MSC-EDKELM-DFS. (j) MSC-EDKELM-WOLFS. Best zoomed-in view.

to train the networks, and the generated disjoint training and test scenes of Indian Pines data set are illustrated in Fig. 17. We can find that the training set selected by CRSS is concentrated locally and dispersed globally, which will greatly reduce the overlap between the superpixels containing the training and test samples than the traditional random sampling strategy. As a result, more realistic performance and more accurate measurements of the real generalization ability of the proposed model can be ensured [13]. The simulations are executed ten times and the concrete average classification results are reported in Table IX. It is observed that our proposed method still achieves the most outstanding performance in terms of AA, OA and kappa coefficient. This demonstrates that the proposed frameworks can extract more distinguished spectral–spatial features, even under the severe disjoint training-test scenes. In contrast, the competitive MSCNN and DR-CNN obtain unsatisfactory accuracies since the patch-wise spatial features cannot play a that significant role without knowing the spatial distribution of each class [52].

F. Ablation Study on SP and ANSP

In our proposed context-aware multiscale perception module, both SP and ANSP are presented to adaptively exploit the local and global contextual information. In order to certify the effectiveness of the SP and ANSP, ablation experiments

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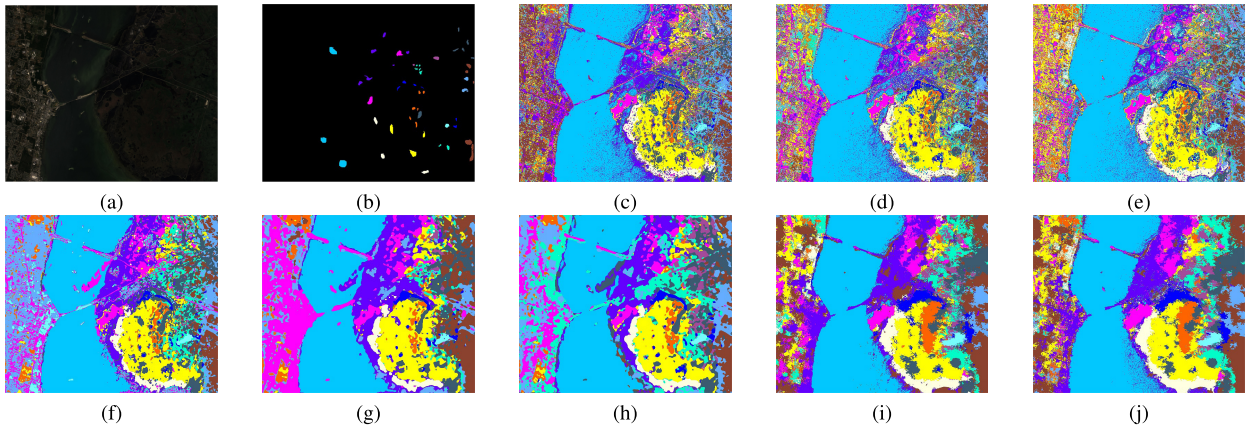

Fig. 14. Classification maps obtained via various algorithms for KSC data set. (a) False-color image. (b) Groundtruth. (c) SVM-RBF. (d) KELM. (e) DKELM. (f) CNN-PPF. (g) MSCNN. (h) DR-CNN. (i) MSC-EDKELM-DFS. (j) MSC-EDKELM-WOLFS. Best zoomed-in view.

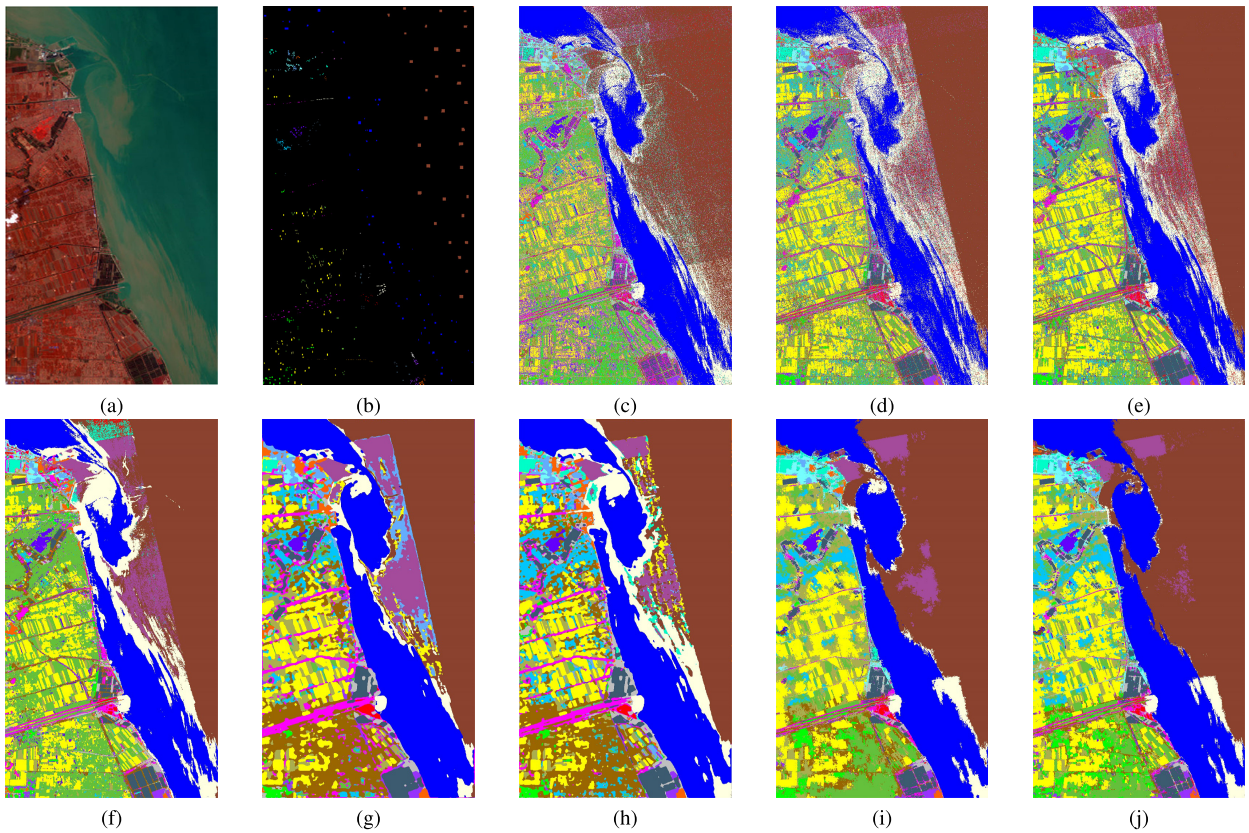

Fig. 15. Classification maps obtained via various algorithms for Yancheng data set. (a) False-color image. (b) Groundtruth. (c) SVM-RBF. (d) KELM. (e) DKELM. (f) CNN-PPF. (g) MSCNN. (h) DR-CNN. (i) MSC-EDKELM-DFS. (j) MSC-EDKELM-WOLFS. Best zoomed-in view.

are carried out in this section. More precisely, the investigated features of the proposed framework can be separated into three folds, i.e., spectral feature, SP and ANSP feature. Therefore, we test the proposed MSC-EDKELM-WOLFS with permutation and combination of these features. Four different cases are listed in the first row of Table X. Additionally, Table X also provides the detailed classification performance on all of the experiment data sets. From the results, we can find that the proposed method with all features always works best for all data sets. Specifically, the OA reduce 7.52%, 3.01% for Indian Pines without SP and ANSP, respectively; 2.31%, 4.17% for University of Pavia without SP and ANSP, respectively; 0.47%, 0.12% for KSC without SP and ANSP, respectively; 1.98% 0.36% for Yacheng data set without SP and ANSP, respectively. This indicates that the SP and ANSP are complementary to jointly learn local-global representations of each pixel. Additionally, if only the spectral feature is left out, the classification performance decreases marginally for University of Pavia and KSC data set, even not cut down for Yancheng data set. It implies that the spectral information has

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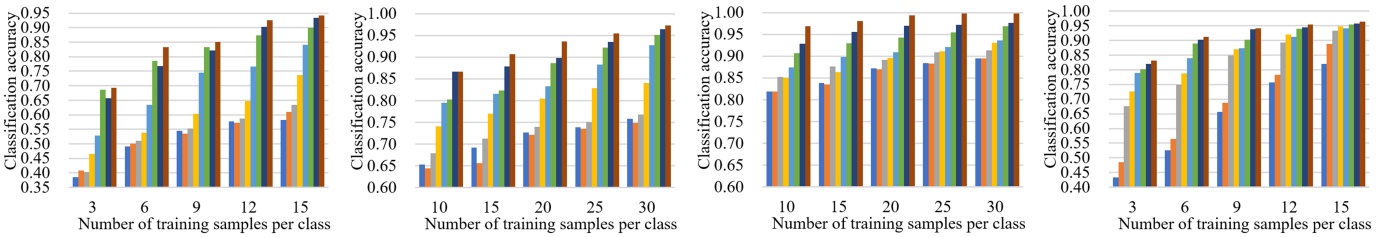
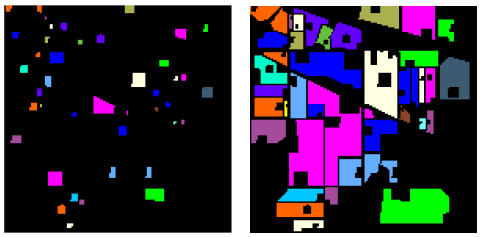


Fig. 16. OA of SVM-RBF, KELM, DKELM, CNN-PPF, MSCNN, DR-CNN, and the proposed MSC-EDKELM-DFS and MSC-EDKELM-WOLFS with different size of training samples. (a) Indian Pines. (b) University of Pavia. (c) KSC. (d) Yancheng.

TABLE IX
CLASSIFICATION ACCURACY OF DIFFERENT ALGORITHMS FOR INDIAN PINES DATA SET UNDER CRSS

ID	training	test	SVM-RBF	KELM	DKELM	CNN-PPF	MSCNN	DR-CNN	MSC-EDKELM-DFS	MSC-EDKELM-WOFLS
C01	5	41	63.41	58.54	56.10	75.61	4.88	43.90	**97.56**	**97.56**
C02	143	1285	66.69	71.60	70.58	80.86	73.70	55.18	**84.90**	84.82
C03	84	746	62.73	58.58	65.55	81.23	65.55	78.95	**90.35**	90.08
C04	24	213	59.15	51.64	58.69	**75.59**	63.38	34.74	77.46	77.46
C05	49	434	86.87	87.33	81.80	94.01	90.55	92.17	**99.54**	**99.54**
C06	73	657	86.91	89.35	88.43	97.26	91.63	90.72	97.41	**98.63**
C07	3	25	88.00	88.00	92.00	48.00	40.00	84.00	**96.00**	**96.00**
C08	48	430	53.49	39.53	44.65	56.05	77.21	81.40	**99.77**	**99.77**
C09	2	18	22.22	27.78	33.33	0.00	0.00	0.00	**100.00**	**100.00**
C10	98	874	46.22	49.20	51.03	62.13	56.64	57.78	62.36	**64.42**
C11	245	2210	68.14	71.72	**75.79**	64.43	54.39	66.52	59.41	64.71
C12	60	533	54.60	52.72	50.84	73.92	65.29	64.92	79.55	**90.43**
C13	21	184	90.22	94.02	97.28	93.48	**100.00**	99.46	99.46	99.46
C14	126	1139	95.17	96.49	96.40	98.86	98.68	98.16	99.65	**99.82**
C15	39	347	43.52	53.60	45.53	49.28	68.01	68.88	65.71	**93.37**
C16	9	84	83.33	82.14	76.19	79.76	97.62	91.67	**100.00**	**100.00**
AA (%)			66.92	67.01	67.76	70.65	65.47	69.28	88.07	**91.00**
OA(%)			68.90	70.25	71.50	76.29	71.38	72.61	80.53	**83.74**
$k \times 100$			64.60	65.99	67.35	73.14	67.51	68.66	77.98	**81.60**



(a) (b)

Fig. 17. Disjoint training-test scenes of Indian Pines data set generated by the CRSS.

been ideally involved in the SP and ANSP features. Since the spectral branch is effective for Indian Pines data set and less-cost in process, we recommend to reserve it in the frameworks.

TABLE X
CLASSIFICATION PERFORMANCE OF THE ABLATION STUDIES FOR SP AND ANSP OF OUR PROPOSED CONTEXT-AWARE MULTISCALE PERCEPTION MODULE

	Features	spectral +SP	spectral +ANSP	SP+ANSP	spectral +SP+ANSP
IndianP	AA(%)	93.16	93.91	95.38	**96.64**
	OA(%)	86.70	91.21	91.27	**94.22**
	$k \times 100$	84.98	89.97	90.07	**93.41**
PaviaU	AA(%)	95.52	91.75	96.39	**96.62**
	OA(%)	94.99	93.13	97.17	**97.30**
	$k \times 100$	93.35	90.91	96.24	**96.42**
KSC	AA(%)	99.24	99.65	99.72	**99.80**
	OA(%)	99.34	99.69	99.75	**99.81**
	$k \times 100$	99.26	99.65	99.72	**99.79**
Yancheng	AA(%)	93.77	94.68	**95.00**	**95.00**
	OA(%)	94.39	96.01	**96.37**	**96.37**
	$k \times 100$	93.21	95.14	**95.58**	**95.58**

G. Discussion on Number of Scales and Kernel Layers

In this section, we investigate the influence of different number of multiple scales and kernel layers, the training sample numbers are same as Tables V–VIII in the experiments. In the multiscale superpixel segmentation stage, the superpixel number in each scale is determined by 11. Actually, with different number of scales, diverse SP and ANSP are generated by the followed context-aware multiscale perception module, which indirectly affect the final classification accuracy. Therefore, we test the proposed MSC-EDKELM-WOLFS with different number of scales, of which the OA is shown in Fig. 18. It is

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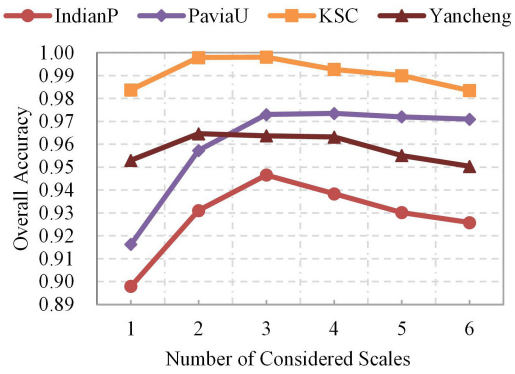

Fig. 18. OA obtained with different number of scales for the experimental data sets.

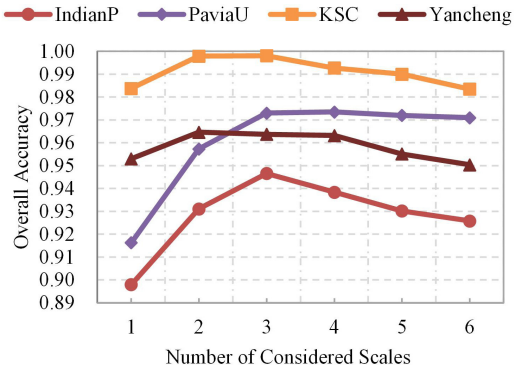

Fig. 19. OA obtained with different numbers of kernel layers in the ensemble DKELM for the experimental data sets.

observed that the performance is enhanced as the number of scales increasing because more useful contextual information is involved into the networks. However, while the scale number becomes bigger, the classification performance decreases. The reason is that under this circumstance, the under-segmentation superpixel maps are produced, which probably lead to a noisy classification result. According to the experimental results, we recommend to utilize three scales to explore the spectral–spatial information.

Furthermore, Fig. 19 shows the OA obtained with different numbers of kernel layers in the ensemble DKELM. It can be seen that the classification performance of EDKELM can reach the optimal result with three kernel layers. This is because, based on the characteristics of hyperspectral data sets, three kernel layers of the network can already extract sufficient refined and distinguished features for the classification task. Whereas, over-fitting occurs when the network is too deep with insufficient training samples.

H. Analysis of the Computational Efficiency

One of the most significant advantages of the presented framework is the high computational efficiency. Concretely, the execution time of the proposed MSC-EDKELM-WOLFS and the compared spectral–spatial approaches, i.e., CNN-PPF, MSCNN, DR-CNN, are reported in Table XI. In our experiments, the hardware equipment is composed of an Intel Xeon E5-1620 v4 processor with 10 M of cache, and a NVIDIA GeForce RTX 2080 GPU. Additionally, MATLAB and Python 3.6 are used as programming languages. It should be noted that the execution time of MSC-EDKELM-WOLFS includes multiscale superpixel segmentation, context-aware multiscale perception and the elapsed time of training and test processes.

TABLE XI
EXECUTION TIME (IN SECONDS) OF THE FOUR EXPERIMENTAL DATA SET

	IndianP	PaviaU	KSC	Yancheng
DKELM	0.81	3.13	0.79	0.84
CNN-PPF	376.48	465.87	545.37	517.23
MSCNN	1243.69	1687.43	1076.65	1439.35
DR-CNN	2380.54	3058.33	2276.69	2846.58
MSC-EDKELM -WOFLS	2.14	12.69	13.78	46.28

From Table XI, it can be seen that DKELM is the fastest method but its performance is not satisfied according to Tables V–VIII. Additionally, we can easily find that the CNN-based deep learning methods are much more time-consuming than the proposed deep networks. In perspective of classification accuracy, DR-CNN can achieve competitive performance. However, due to the procedure of calculating diverse convolutional regions and training the networks, its computational complexity is the highest. Approximately, it spends 61 to 1112 times than that of the proposed MSC-EDKELM-WOFLS in different data sets. In addition, it is worth mentioning that all portions of MSC-EDKELM, i.e., multiscale superpixel segmentation, context-aware multiscale perception module and ensemble DKELM are performed independently on each considered scale, which means that the proposed frameworks can potentially go parallel. Therefore, the complement of our proposed method can be further accelerated.

V. CONCLUSION

In this paper, we propose a novel MSC-EDKELM to realize accurate and efficient HSI classification. First, through using adaptive multiscale superpixel segmentation approach, diverse local and global contextual information is precisely captured by the SP and ANSP, which is proven to be conducive to classify the boundary pixels in our experiments. Second, the proposed ensemble DKELM is able to simultaneously extract the high-level spectral–spatial features, which demonstrates excellent generalization ability with different number of training samples, even under the difficult SSS conditions and disjoint training-test scenes. Particularly, the experimental results also indicate that the well-designed WOLFS is powerful to take the advantages of each independent DKELM classifier. Most importantly, the proposed method can be implemented with a relatively fast speed, which is critical for real-time applications.

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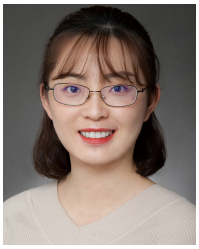

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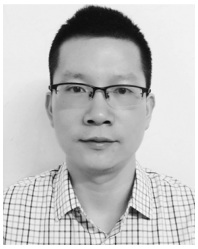

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